

Book 7 — POP Osnias server & private Osnias Clearing network

Title	Book 7 — O-POP Osnias server & private Osnias Clearing network
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Executive summary

Explain the architectural choices adopted for the Osnias Mainnet infrastructure: functional separation, private RPC access to the Sei network, VPN access for end users, dedicated netting computation, Ethereum escrow read access, traceability and resilience.

Keywords

Osnias POP; Osnias Clearing; private Sei RPC; Osnias-Core; clearing; Sei; Ethereum

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I Position within the documentation corpus and status

Book 7 is a standalone document forming part of the logical continuity of the Osnias Clearing documentation corpus.

Document	Function	Contribution to the onboarding journey
Book 1	Osnias-ID	Participant identity and cross-chain linkages.
Book 2	Architecture	Separation of clearing, settlement and collateral.
Book 3	Clearing Rules	Obligations, netting, governance and responsibilities.
Book 4	Cycles and Calendar	Time synchronization and transaction states.
Book 5	Security	Domains A, B and C, composite condition and threat model.
Book 6	Onboarding	Admission, EUR/EURC gateway, escrow and exit.
Book 7	POP Osnias	Describes a server node providing access to the Osnias network.

Document status This document describes a functional architecture and design requirements. It does not constitute an independent audit, formal verification, penetration test, security certification or authorization for production deployment.

II Purpose of the architecture

The Osnias POP server and network architecture is designed around a simple principle: critical clearing functions must not depend on the same resources as user access, nor on a third-party public RPC endpoint. Each layer has an explicit and measurable role.

- Sei is the canonical ledger for clearing positions and events.
- Ethereum provides the escrow / collateral and settlement layer associated with the relevant accounts.
- Osnias-Core performs netting calculations, node-level aggregation and reconciliation controls.
- The Osnias POP provides network access, monitoring, logging and operational resilience.
- End users access a dedicated Osnias RPC network through an authenticated VPN. This network is separate from the Administration LAN and provides no access to maintenance or monitoring interfaces.

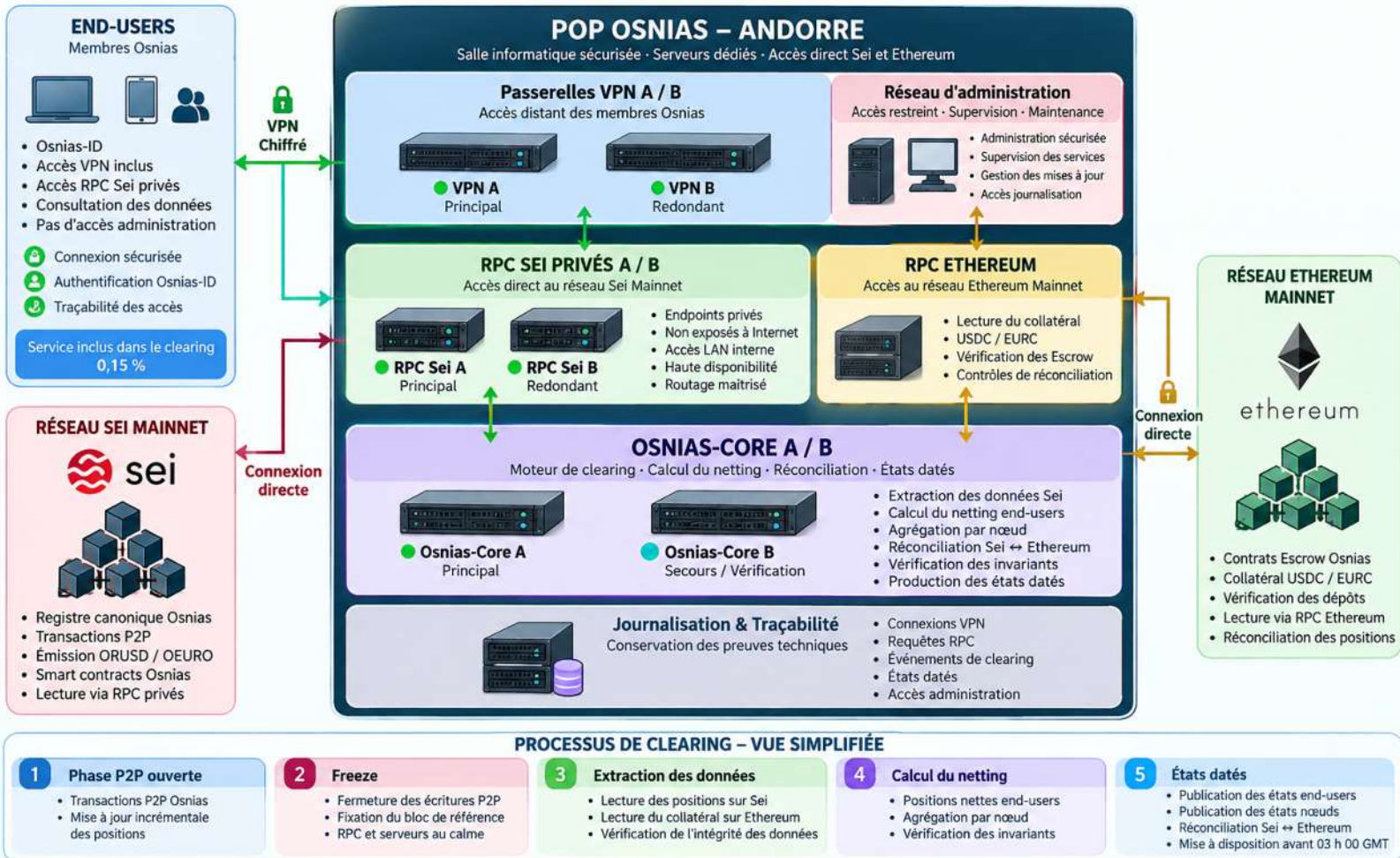
Guiding principle

Separating the ledger, computation, network access and escrow layers allows each function to be sized, secured and audited independently.

III Topology of the OSNIAS network and an O-POP

OSNIAS CLEARING – ARCHITECTURE SERVEUR MAINNET

Infrastructure sécurisée · Salle blanche · Andorre



Companion document to the “Osnias Clearing — Mainnet Server Architecture” diagram.

IV Why dedicated private Sei RPC servers?

An RPC server does not constitute a separate blockchain network: it provides a technical interface between Osnias applications and the Sei network. The private RPC server remains connected to Sei Mainnet, follows the chain, verifies it and exposes the data required by authorized applications. The distinction lies in control of RPC access, not in the public or canonical nature of the blockchain.

IV.b What “private RPC” means

- The RPC endpoint is not exposed to the public Internet as an open service.
- Requests from Osnias members pass through the VPN and defined authorization rules.
- Osnias-Core servers use an internal network path and do not pass through the end-user VPN gateway.
- The RPC server remains connected to the peers / services required to follow the Sei network and its canonical state.

Component	Private / controlled	Remains public / external
Osnias end-user RPC network	Dedicated network: VPN access, authentication, quotas and logging	—
Administration LAN	Private local Osnias network, 10.x.x.x addressing, administration only	—
Sei blockchain	—	Consensus, blocks and canonical network state
Internet traffic	Filtered at the POP	Upstream routing not fully controllable

IV.c Why not rely solely on public RPC endpoints?

Public or shared endpoints are useful for development and as a fallback, but they introduce operational dependencies: quotas, saturation, routing changes, third-party maintenance or rate-limiting policies. For clearing infrastructure, Osnias seeks to control the access path to critical data and to measure its availability precisely.

Intended effect

Reduce dependency on public RPC infrastructure and third-party provider rerouting policies while maintaining normal connectivity to Sei Mainnet.

V RPC / Osnias-Core separation

RPC servers and computation servers perform different functions. RPC servers follow the chain and provide data; Osnias-Core applies the clearing logic. Separating them prevents netting computation from directly competing with indexing and RPC service workloads.

Layer	Primary functions	What it must not do
RPC Sei A/B	Chain tracking, state reads, RPC exposure, availability monitoring	Run netting business logic

Osnias-Core A/B	Consolidation, netting, node-level aggregation, invariant controls, timestamped states	Serve as a public RPC endpoint
VPN A/B	Encrypted transport and member access	Directly administer Core servers
Logging	Retain technical logs and operational events	Modify clearing state

VI Cycle-oriented architecture: P2P open, then freeze

The protocol's time-based logic allows the infrastructure to operate differently depending on the cycle state. While P2P is open, Osnias-Core can maintain an incremental projection of positions. At freeze, new P2P writes are suspended at protocol level and a reference state is fixed.

- 1 — P2P open** — Ingestion of Sei events and incremental position updates.
- 2 — Freeze** — Stop new P2P obligations and fix the reference block / state.
- 3 — Consolidation** — Verify that expected events have been incorporated and are consistent.
- 4 — Netting** — Calculate end-user net positions, then aggregate by node.
- 5 — Controls** — Invariants, Sei <-> Ethereum reconciliation, discrepancy detection.
- 6 — Timestamped states** — Produce and publish states according to the Osnias operational schedule.

Consequence

The freeze does not “make the servers inactive”: it stops the creation of new P2P obligations and allows critical capacity to be reserved for consolidation, netting and controls.

VII Sei <-> Ethereum reconciliation

Netting computation is based on clearing positions recorded on Sei. Reading the Ethereum escrow provides a separate coverage and reconciliation control. Osnias-Core reconciles both states at a common logical cut-off, taking into account the finality rules specific to each network.

Source	Data	Use	Control
Sei	End-user clearing position / balance	Netting and clearing state	Consistency with the reference block
Ethereum	Eligible amount locked in Escrow	Collateral coverage / availability	Contract, account, finality and escrow state

Osnias-Core	Calculated discrepancy between relevant states	Anomaly detection	Invariant rules and logging
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Where Osnias rules require post-clearing equality between the compared values, the control may be expressed as:

$$\Delta u = \text{EligibleEscrowEthereum}(u) - \text{ClearingStateSei}(u)$$

Expected state at the defined control point: $\Delta u = 0$

This equality is not presented as a permanent identity at every moment: it applies at the control point defined by the clearing rules and after logical synchronization of the two sources.

VIII End-user VPN: service access without exposing RPC infrastructure

VPN access is included in the Osnias service for authorized end users. Its purpose is to provide an encrypted path to the dedicated private RPC network and to enable an identifiable, traceable and revocable access policy.

- The Osnias-ID serves as the member identification reference within the Osnias system; it must not be used by itself as an authentication secret.
- VPN authentication must rely on distinct and revocable cryptographic mechanisms.
- End users have no access to the Administration LAN or maintenance interfaces.
- Quotas and service classes may preserve capacity for critical functions during clearing windows.

Security

The VPN protects the path between the user and the POP; it does not make Sei transactions private and does not alter the visibility inherent to the public ledger.

IX Redundancy and service continuity

A/B redundancy is intended to prevent a single failure from interrupting service. It must be designed by failure domain: computation, RPC, VPN, network, power, storage and upstream connectivity.

- Sei RPC A/B: continuity of access to network state and availability comparison.
- Osnias-Core A/B: primary and standby / verification servers, with controls preventing duplicate publication.
- VPN gateways A/B: continuity of access for authorized members.
- Redundant network switching and power: reduction of local points of failure.
- Separate logging: preservation of records even during an application service incident.
- Tested failover procedures: hardware redundancy has value only if recovery is measurable and repeatable.

Two devices located in the same room may remain exposed to a common cause of failure. The Mainnet architecture must therefore distinguish local high availability from geographic disaster recovery, according to the service level ultimately adopted.

X Security, neutrality and availability of the Osnias POP

The security, operational neutrality and availability of an Osnias POP are vital parameters of the architecture.

The POP must be designed to avoid any single point of failure capable of interrupting access to the ledger, computation functions, reconciliation or services required for clearing. Critical components are therefore deployed

using a redundant architecture, including Sei RPC servers, Osnias-Core servers, VPN gateways, network equipment, power supply and logging functions.

The architectural objective is continuous service availability, including in the event of failure of an individual server or device. Failover between redundant components must be controlled, tested and traceable. This objective is not an absolute guarantee of 100% availability, but a design requirement based on redundancy and the absence of a single point of failure.

The RPC network for end users is a dedicated network, separate from the Administration LAN. The Administration LAN uses a private IPv4 address range in 10.x.x.x, reserved for local operations, monitoring and maintenance communications. It is neither directly exposed to the Internet nor accessible from the end-user RPC network.

Physical co-location of these infrastructures in the same secure room or data center does not compromise this separation: network domains, access rights, interfaces and functions remain logically and technically distinct.

POP neutrality requires controlled, traceable and resilient access to the Sei network and Osnias functions, without introducing any discretionary mechanism capable of altering, filtering or prioritizing clearing states outside the protocol rules.

XI Why dedicated netting computation servers?

Netting computation is a deterministic, parallelizable workload that is distinct from the operation of an RPC node. Dedicated servers allow CPU, memory and storage resources to be reserved for consolidation operations without disrupting access to the ledger.

- Incremental pre-computation of positions during the P2P phase.
- Local / private reading of Sei data at freeze time.
- Parallel account processing and node-level aggregation.
- Reconciliation with Ethereum escrow states.
- Invariant controls executed before publication of timestamped states.

The expected performance gain may be significant compared with a full reconstruction from history at each cut-off. It must nevertheless be quantified through performance testing: extraction time, computation time, reconciliation time and total time to publication.

XII Traceability: what the POP must evidence

POP logging is intended to reconstruct the technical execution of a cycle and access to the service. It does not replace on-chain evidence; it complements it with operational evidence.

- VPN connection and disconnection, technical identity and associated rights.
- Relevant RPC requests, errors, quotas and latency.
- Reference block / state used for a cut-off.
- Computation engine version and cycle parameters.
- Invariant and reconciliation results.
- A/B failover, incidents, recovery and administrative interventions.

Important distinction

An RPC log demonstrates that a service received and processed a request. Evidence of canonical state and blockchain execution must remain based on network data and the selected finality criteria.

XIII Target architecture summary

Component	Role in the architecture
Secure server room / POP	Controlled hosting of the Osnias Mainnet infrastructure.
RPC Sei A/B	Private and redundant access to the canonical Sei ledger, connected to the public Sei network.
Osnias-Core A/B	Netting computation, aggregation, invariants, reconciliation and timestamped states.
RPC / Ethereum access	Reading escrow contracts and collateral controls.
VPN A/B	Encrypted member access to authorized POP services.
Administration LAN	Private local 10.x.x.x network reserved for operations, monitoring and maintenance; separated from the end-user RPC network.
Logging	Technical traceability independent of computation functions.
Freeze-aware scheduling	Prioritization of critical resources at cut-off and during netting.
End-user RPC network	Dedicated network for VPN access to authorized RPC services; separate from the Administration LAN even when co-located in the same data center.

XIV Conclusion

The purpose of dedicated infrastructure is not to create a private network competing with Sei. Rather, it provides Osnias with a controlled access point to Sei Mainnet while clearly separating RPC access, clearing computation, Ethereum escrow and end-user access.

This architecture reserves computing resources for critical phases, reduces operational dependency on public endpoints, strengthens traceability and prepares for measurable Mainnet operations. It is a target architecture: its capacity, availability objectives and performance must be demonstrated before production deployment.

Summary statement

Sei records. Ethereum escrows. Osnias-Core computes and reconciles.
The POP secures, distributes and traces access to the infrastructure.